

CO2 Question Bank

1. Show that the language $L = \{ 0^n 1^n / n \geq 1 \}$ is not regular.
2. Show that the language $L = \{ L = \{ a^p \mid p \text{ is a prime number} \} \}$ is not regular.
3. Prove that the language $L = \{ a^n b^n c^n / n \geq 1 \}$ is not regular.
4. Check whether the language $L = \{ WCW^R \mid W \text{ in } \{ 0 + 1 \}^* \}$ is regular or not..
5. Prove that $L = \{ a^{i^2} \mid i \geq 1 \}$ is not regular.
6. Explain parse tree and obtain Left most derivation and right most derivation for the string aaabbabba from the grammar.
 $S \rightarrow aB \mid bA$
 $A \rightarrow a \mid aS \mid bAA$
 $B \rightarrow b \mid bS \mid aBB$
7. Consider the grammar, (i) $S \rightarrow iCtS \mid iCtSeS \mid a$, $C \rightarrow b$, Where i, t, and e stand for if, then, and else, and C and S for “conditional” and “statement” respectively. S is the start symbol then
 - (i) Construct a leftmost derivation for the sentence $w = ibtibtaea$.
 - (ii) Show the corresponding parse tree for the above sentence.
 - (iii) Is the above grammar ambiguous? If so, prove it.
8. Show that the CFG G_1 with productions $S \rightarrow aSaSbS \mid aSbSaS \mid bSaSaS \mid \varepsilon$ and CFG G_2 with productions $S \rightarrow aSbS \mid bSbS \mid bSaS \mid \varepsilon$ can generate the same language and Construct its parse trees.
9. Consider the grammar $G = (\{S\}, \{a,b\}, \{S \rightarrow aSb, S \rightarrow ab\}, S)$. Find the language generated by G.
10. Find the grammar for the language consist the strings of palindrome.
11. Given the grammar $G = (\{E\}, \{id\}, \{E \rightarrow E+E \mid E * E \mid id\}, \{E\})$. Derive the string $id + id * id$ using left most derivation, right most derivation and draw parse tree
12. Show that the following grammar is ambiguous with respect to the string aaabbabbbba
 $S \rightarrow aB \mid bA$
 $A \rightarrow a \mid aS \mid bAA$
 $B \rightarrow b \mid bS \mid aBB$
13. **Demonstrate** the grammar is ambiguous $S \rightarrow SS \mid (S) \mid S(S)S \mid e$
14. Let G be the grammar $S \rightarrow 0B \mid 1A$, $A \rightarrow 0 \mid 0S \mid 1AA$, $B \rightarrow 1 \mid 1S \mid 0BB$. For the string 00110101 find (i) Leftmost Parse tree
 (ii) Rightmost Parse tree
15. What is ambiguity in grammar? Show that the grammar $G = (\{S\}, \{a,b, \varepsilon\}, P, S)$ with production P as $S \rightarrow aS \mid aSbS \mid \varepsilon$ is ambiguous.
16. Show that the grammar $G = (\{S,A\}, \{0,1, \varepsilon\}, P, S)$ with Production P as $S \rightarrow AS \mid \varepsilon$;
 $A \rightarrow A1 \mid 0A1 \mid 01$ is ambiguous
17. Show that the grammar $S \rightarrow SbS \mid a$ is ambiguous. Show that the string ‘abababa’ has two
 - (a) Parse tree
 - (b) Leftmost Derivation
 - (c) Rightmost Derivation
18. Let G be the grammar $E \rightarrow E+T \mid T$, $T \rightarrow T * F \mid F$, $F \rightarrow (E) \mid a \mid b \mid c$ Show that the string $(a+b+c*a)$ is derived using LMD and RMD
19. Let G be the grammar $S \rightarrow A1B$ $A \rightarrow 0A \mid \varepsilon$ $B \rightarrow 0B \mid 1B \mid \varepsilon$ Show that the grammar is Unambiguous

20. Construct CFG for the language $L = \{w \in \{(,)\}^* / w \text{ is balanced and structured}\}$. The language consists of strings over '(' and ')'. Example string for balanced and structured: (()) ().
21. Find CNF for the grammar $S \rightarrow ASB, A \rightarrow aASA \mid a \mid \epsilon, B \rightarrow SbS \mid A \mid bb$.
22. Find GNF for the grammar $G = (\{A_1, A_2, A_3\}, \{a, b\}, P, A_1)$ where P is defined as $A_1 \rightarrow A_2A_3, A_2 \rightarrow A_3A_1 \mid b, A_3 \rightarrow A_1A_2 \mid a$.
23. Show that the CFG G_1 with productions $S \rightarrow aSaSbS \mid aSbSaS \mid bSaSaS \mid \epsilon$ and CFG G_2 with productions $S \rightarrow aSbS \mid bSbS \mid bSaS \mid \epsilon$ can generate the same language.
24. Consider the following grammar for list structures: $S \rightarrow a \mid \wedge \mid (T), T \rightarrow T, S \mid S$. Find left most derivation and parse tree for the string $((a, a), \wedge(a)), a$. Also draw its derivation tree.
25. Write context free grammar for the language $L = \left\{ \frac{ww^R}{w} \in \{a, b\}^* \text{ and } w^R \text{ is the reverse of the string } w \right\}$
26. Show that the following context free grammar is ambiguous.
 $S \rightarrow a \mid Sa \mid bSS \mid SSb \mid SbS$
27. Simplify the following grammar and convert to Chomsky Normal Form
 $S \rightarrow ASB \mid \epsilon$
 $A \rightarrow aAS \mid a$
 $B \rightarrow SbS \mid A \mid bb$
28. Convert the following grammar into GNF $S \rightarrow AA/a, A \rightarrow SS/b$
29. Convert the grammar
 $S \rightarrow AB$
 $A \rightarrow BS \mid b$
 $B \rightarrow SA \mid a$ into GNF
30. **Give** the Chomsky normal form equivalent to the grammar
 $S \rightarrow AB \mid aB,$
 $A \rightarrow bAA \mid aSa \mid a$
 $B \rightarrow aBB \mid bS \mid b$
31. Construct a context free grammar for the following language
 $L = \{\text{set of all strings of } |w|=2 \text{ over } \{a, b\}\}$
 $L = \{\text{set of all strings of } |w| \geq 2 \text{ over } \{a, b\}\}$
32. Convert the following grammar $G = (\{A, B, C, D, Y\}, \{a, b, c, d, g, f\}, P, S)$ with Production P as
 $A \rightarrow BCD \mid b; B \rightarrow Yc \mid d; C \rightarrow gA \mid c; D \rightarrow dB \mid a; Y \rightarrow f$ into Greibach Normal Form.
33. Design a context free grammar for the language 1. $L = \{w \mid w \in \{a, b\}^*, \#_a(w) = \#_b(w) \text{ where } \#_i(w) \text{ is the number occurrences of a symbol } i \text{ in a string } w\}$.